Medical Terminology Identification Assignment

Oscar Cortez
AI in Healthcare (ITAI-4375)
Dr. Lina Awodogan
January 21, 2026

Introduction

Medical terminology, often referred to as *termini technici*, forms the foundation of effective communication in medicine and healthcare, particularly as artificial intelligence (AI), machine learning (ML), and deep learning (DL) become increasingly integrated into clinical practice. These technologies depend on precise, standardized medical language to accurately analyze clinical data, support decision-making, and improve patient outcomes. A strong understanding of medical terminology is therefore essential for healthcare professionals and AI-driven systems involved in diagnosis, predictive analytics, treatment planning, and population health management.

Identified Medical Terms

Term 1: Myocardial Infarction

Definition: The death of heart muscle tissue caused by prolonged lack of blood supply (Centers for Disease Control and Prevention [CDC], 2023).
Context: "Patients presenting with acute chest pain were evaluated for myocardial infarction using clinical and diagnostic criteria" (Kim et al., 2023).
Importance: Understanding this term is crucial for recognizing and treating heart attacks, a leading cause of morbidity and mortality worldwide.

Term 2: Hypertension

Definition: A chronic condition characterized by persistently elevated arterial blood pressure (World Health Organization [WHO], 2023).
Context: Hypertension is frequently identified as a major cardiovascular risk factor in emergency and population health studies (Mills et al., 2022).
Importance: Early identification and management of hypertension reduces the risk of stroke, heart disease, and other complications.

Term 3: Sepsis

Definition: A life-threatening organ dysfunction caused by a dysregulated immune response to infection (Rhee et al., 2023).
Context: Recent studies describe the use of AI-based early warning systems to detect sepsis before clinical deterioration occurs (Henry et al., 2022).
Importance: Early recognition of sepsis is critical in both pre-hospital and hospital settings to reduce mortality.

Term 4: Neoplasm

Definition: An abnormal growth of tissue that may be benign or malignant (National Cancer Institute [NCI], 2024).
Context: Neoplasms are commonly discussed in oncology reports related to cancer screening and diagnostic imaging (Siegel et al., 2024).
Importance: Understanding neoplasms is essential for accurate cancer diagnosis, staging, and treatment planning.

Term 5: Ischemia

Definition: A condition in which there is insufficient blood flow to an organ or tissue (American Heart Association [AHA], 2023).
Context: Ischemia is frequently referenced in stroke and cardiovascular emergency research (Powers et al., 2023).
Importance: Prompt recognition of ischemia can prevent irreversible tissue damage and improve patient outcomes.

Term 6: Metastasis

Definition: The spread of cancer cells from a primary tumor to distant organs (NCI, 2024).
Context: Metastasis is a central focus in studies examining cancer progression and survival outcomes (Sung et al., 2021).
Importance: The presence of metastasis significantly affects prognosis and treatment decisions.

Term 7: Hypoxia

Definition: A condition in which tissues are deprived of adequate oxygen supply (National Institutes of Health [NIH], 2023).
Context: Hypoxia is commonly discussed in pulmonary and critical care research, particularly in respiratory failure cases (Mellado-Artigas et al., 2021).
Importance: Monitoring hypoxia is vital in emergency and critical care to prevent organ damage and death.

Term 8: Biomarker

Definition: A measurable indicator of a biological or pathological process or response to treatment (FDA, 2022).
Context: Biomarkers are widely used in precision medicine and AI-driven diagnostic research (Topol, 2023).
Importance: Biomarkers enable early disease detection and personalized treatment strategies.

Term 9: Comorbidity

Definition: The presence of one or more additional diseases alongside a primary condition (Centers for Medicare & Medicaid Services [CMS], 2022).
Context: Comorbidities are frequently analyzed in epidemiological studies assessing patient outcomes (Valderas et al., 2021).
Importance: Comorbidities complicate clinical management and are critical variables in predictive healthcare models.

Term 10: Telemedicine

Definition: The delivery of healthcare services remotely through digital communication technologies (U.S. Department of Health & Human Services [HHS], 2023).
Context: Telemedicine use expanded significantly in recent healthcare policy and technology reports (Koonin et al., 2022).
Importance: Telemedicine improves access to care and relies heavily on AI for triage, monitoring, and follow-up services.

Conclusion

Medical terminology plays a critical role in ensuring accurate communication, diagnosis, and treatment in healthcare. As AI, ML, and DL continue to shape modern medicine, precise use of medical terms supports reliable data analysis and clinical decision-making. Mastery of medical terminology benefits both healthcare professionals and AI systems, ultimately improving patient outcomes across emergency, clinical, and population health settings.

References

American Heart Association. (2023). *What is ischemia?* https://www.heart.org

Centers for Disease Control and Prevention. (2023). *Heart disease facts*. https://www.cdc.gov

Food and Drug Administration. (2022). *Biomarker qualification program*. https://www.fda.gov

Henry, K. E., et al. (2022). A targeted real-time early warning score (TREWScore) for septic shock. *Science Translational Medicine, 14*(630), eabm4760.

Kim, Y., Lee, J., Kim, H., Shin, D. W., & Cho, B. (2023). Association between social activity and awareness of warning symptoms of stroke and myocardial infarction. *Frontiers in Public Health, 10*, 1058871.

National Cancer Institute. (2024). *Cancer terminology*. https://www.cancer.gov

Rhee, C., Jones, T. M., Hamad, Y., et al. (2023). Prevalence, underlying causes, and preventability of sepsis-associated mortality. *JAMA Network Open, 6*(2), e2253846.

Topol, E. (2023). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine, 29*(1), 44–56.

World Health Organization. (2023). *Hypertension*. https://www.who.int